

## Exam 3 Review Sheet

### Internal Focus:

- Sign conventions (crib sheet)
- Method: "section"

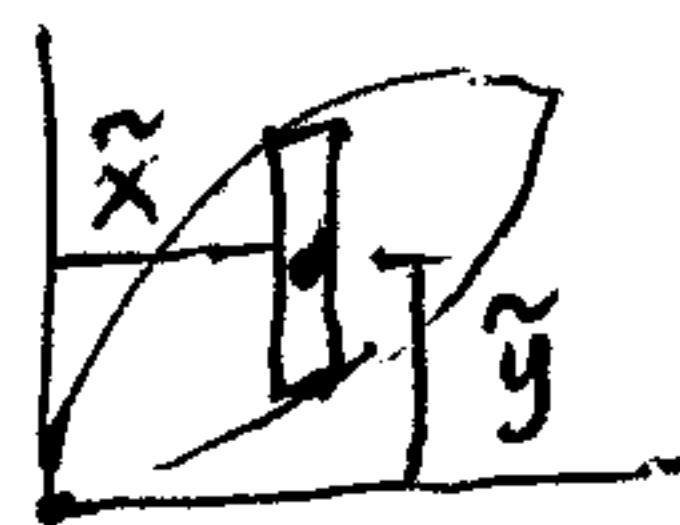
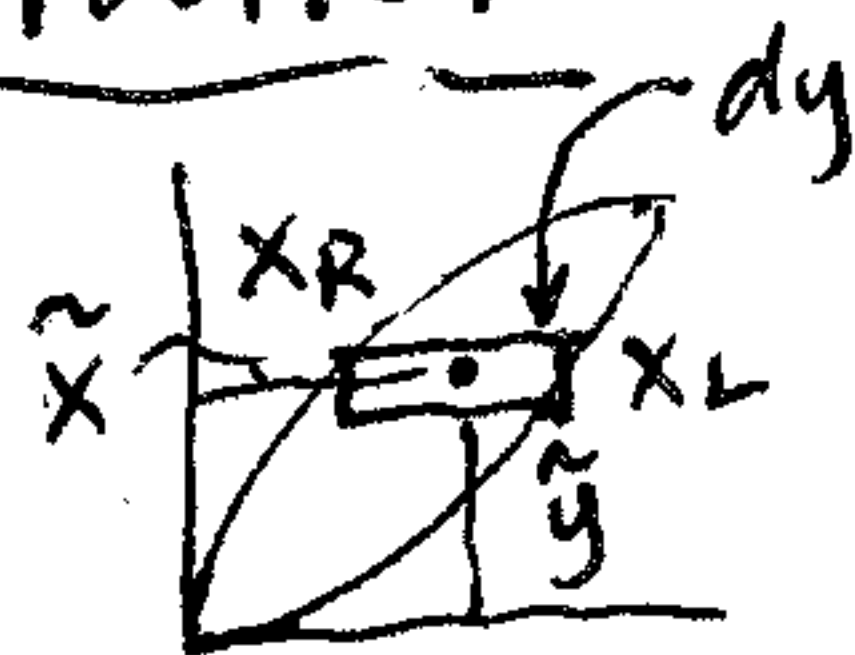
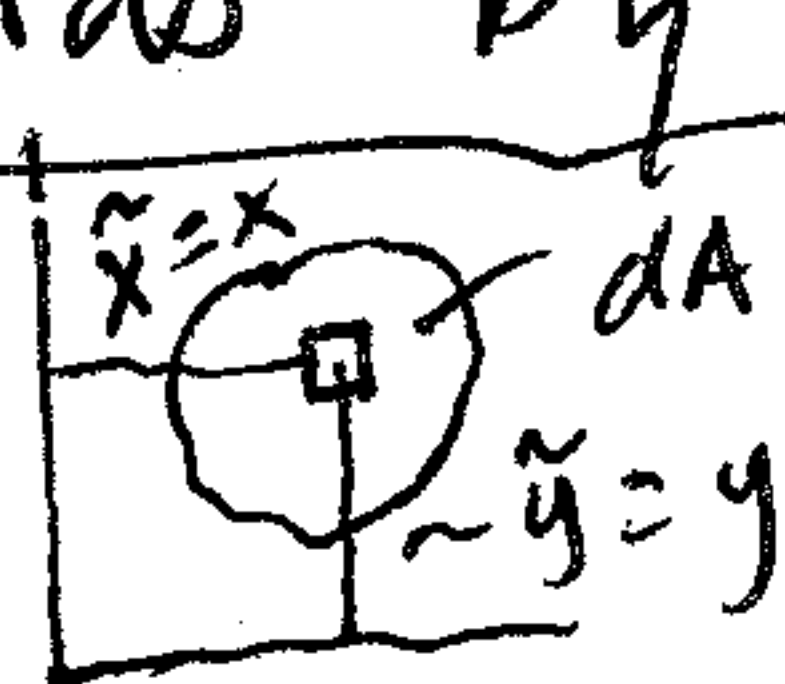
V&M: Know/memorize the rules!

- Equations:

### Friction:

- General Probs
- Belt Friction
- Wedges
- Self-locking condition
- Multi-mode problems  
Check tip/slip
- Do not always set  $F = \mu N$  (only if on verge of motion)
- Sometimes solve for  $F$  and  $N$ , then check:  
 $\frac{F}{N} \leq \mu$ ?

### Centroids by Integration



Centroids: Composite Areas, Lines, Volume

### P&G Problems:

$$A = \sum \theta \bar{r} L$$

$$V = \sum \theta \bar{r} A$$